

COL100: Introduction to Computer Science

Lab 3: Functions

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1 Functions

Functions are a fundamental building block of programming and play a crucial role in organizing code, making it more modular, and enhancing reusability. In this lab, we will explore the concept of functions by working through various sample programs.

1.1 Perimeter of a Triangular Garden

In this sample program, we will explore the usage of functions by creating a program that calculates the perimeter of a triangular garden based on the coordinates of its vertices. The program will use a user function called `distance` to compute the distance between two points in the Cartesian plane.

Next, we provide the implementation or definition of the `distance` function.

```
1 def distance(x1, y1, x2, y2):
2     dx = x2 - x1
3     dy = y2 - y1
4     dist = math.sqrt(dx * dx + dy * dy)
5     return dist
```

The function definition includes its name `distance` and the parameter list `x1`, `y1`, `x2`, and `y2` representing the coordinates of two points in the Cartesian plane. Inside the function, we calculate the distance between two points. Finally, the function returns the computed distance.

We can prompt the user for input, make the function calls to calculate the perimeter and display the result. Here is the complete program that calculates the perimeter of a triangular garden given the coordinates of its vertices. Note that the `distance` function is called three times, each time with a different set of coordinates. The calculated distances are added to obtain the perimeter of the triangular garden.

```
1 import math
2
3 # Function to compute the distance between two points
4 def distance(x1, y1, x2, y2):
5     dx = x2 - x1
6     dy = y2 - y1
7     dist = math.sqrt(dx * dx + dy * dy)
8     return dist
9
```

```

10 # Get coordinates of vertices
11 x1 = float(input("Enter x-coordinate of vertex 1:"))
12 y1 = float(input("Enter y-coordinate of vertex 1:"))
13 x2 = float(input("Enter x-coordinate of vertex 2:"))
14 y2 = float(input("Enter y-coordinate of vertex 2:"))
15 x3 = float(input("Enter x-coordinate of vertex 3:"))
16 y3 = float(input("Enter y-coordinate of vertex 3:"))
17
18 # Compute perimeter
19 perimeter = (distance(x1, y1, x2, y2) +
20             distance(x2, y2, x3, y3) +
21             distance(x3, y3, x1, y1))
22
23 # Display result
24 print(f"Perimeter of triangle = {perimeter}")

```

1.2 Swap Two Numbers

This example illustrates the scope of parameters where the function parameters receive copies of the actual parameters, and any modification made within the function does not impact the original variables outside of it.

```

1 # Function to swap two numbers
2 def swap_numbers(a, b):
3     # Swapping using a temporary variable
4     temp = a
5     a = b
6     b = temp
7     return a, b
8
9 # Input from the user
10 num1 = int(input("Enter the first number: "))
11 num2 = int(input("Enter the second number: "))
12
13 # Print before swapping
14 print(f"Before swapping: First number = {num1}, Second number = {
    num2}")
15
16 # Call the function to swap numbers
17 num1, num2 = swap_numbers(num1, num2)
18
19 # Print after swapping
20 print(f"After swapping: First number = {num1}, Second number = {num2
    }")

```

2 Submission Problem

2.1 Instructions

- Complete the code of the following problem in **main.py** program file.
- Submit the program file in the assignment titled “Lab 3” on Gradescope.
- The Autograder on Gradescope will evaluate the submitted program on some test cases.

2.2 Euclidean Distance between Two Vectors

Write a program to calculate the Euclidean distance between two vectors in 3D space using a function. The Euclidean distance between two vectors $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$ is defined as:

$$d(A, B) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

Please implement your program that satisfies the following specifications.

- Your program should read two vectors, each representing a 3D vector.
- Your program should calculate and print the Euclidean distance (rounded up to two decimal places) between the two vectors.

You can implement a function as follows.

```
1 euclidean_distance(a1, a2, a3, b1, b2, b3);
```

2.2.1 Sample Input 1

```
1 1 2 3
2 4 5 6
```

2.2.2 Sample Output 1

```
1 5.20
```

2.2.3 Sample Input 2

```
1 8 9 7
2 3 2 5
```

2.2.4 Sample Output 2

¹ 8.83

3 Practice Problems

3.1 Cosine Similarity

Write a program to calculate the cosine similarity between two 3D vectors using functions. The cosine similarity is defined as:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

The L2 (Euclidean) norm of a vector V is given by:

$$\|V\| = \sqrt{\sum_{i=1}^3 V_i^2}$$

Please implement your program that satisfies the following specifications:

- Your program should read a positive integer n (which will be fixed at 3 for 3D space).
- Your program should then read 3 floating-point numbers corresponding to the first vector.
- Your program should then read 3 floating-point numbers corresponding to the second vector.
- Your program should print the following four space-separated floating-point numbers:
 - L2 Norm of the first vector
 - L2 Norm of the second vector
 - Dot product of the two vectors
 - Cosine similarity between the two vectors

You can implement the following functions:

- `l2_norm(vector)`
- `dot_product(vector1, vector2)`
- `cosine_similarity(vector1, vector2)`

You can implement the following functions.

```
1 l2_norm(vector , dimension);  
2 dot_product(vector1 , vector2 , dimension);  
3 cosine_similarity(vector1 , vector2 , dimension);
```

3.1.1 Sample Input 1

```
1 2
2 1 0
3 0 1
```

3.1.2 Sample Output 1

```
1 1.00 1.00 0.00 0.00
```

3.1.3 Sample Input 2

```
1 2
2 1 1
3 1 1
```

3.1.4 Sample Output 2

```
1 1.41 1.41 2.00 1.00
```

3.2 Five Numbers and a Cipher

Write a program that will take five numbers which act like a cipher. To unlock the cipher, you need to perform the following operations on the input (a,b,c,d,e) three times.

```
1 a = a - b
2 b = b - c
3 c = c - d
4 d = d - e
5 e = e - a
```

Please implement your program that satisfies the following specifications.

- Your program should read a space separated input of 5 integers.
- Your program should print the values in a single line with space separation for a,b,c,d,e.

You can implement the following functions.

```
1 evaluate(a, b, c, d, e);
```

3.2.1 Sample Input 1

```
1 250 200 150 100 50
```

3.2.2 Sample Output 1

```
1 0 0 -50 50 0
```

Explanation:

The numbers after the first execution of the function are 50 50 50 50 0

The numbers after the second execution of the function are 0 0 0 50 0

The numbers after the third execution of the function are 0 0 -50 50 0

3.3 Characters and Numbers

You are given a 4 letter word as an input in a prescribed format, you are tasked with finding the sum of the characters in this word. You can use the following functions:

- `ord()` converts a character to its corresponding Unicode code point
- `chr()` converts a Unicode code point back to its character.

3.3.1 Sample Input 1

```
1 C O D E
```

3.3.2 Sample Output 1

```
1 283
```

3.3.3 Sample Input 2

```
1 Play
```

3.3.4 Sample Output 2

```
1 406
```